Lab 5: Static Code Analysis

BHUVI BAGGA
PES2UG24CS117
5B

1. Pylint Analysis

```
Barrier Key configured: False
(venv) PS C:\Users\dell\Downloads\L5_51_parking_lot_system-main\L5_51_parking_lot
_system-main> python -m pylint parking_lot_system.py

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 9.71/10, +0.29)
```

2. Bandit Security Analysis

```
(venv) PS C:\Users\dell\Downloads\L5_51_parking_lot_system-main\L5_51_parking_lot
_system-main> python -m bandit -r parking_lot_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.5
Run started:2026-10-14 12:14:27.521011+00:00

Test results:
        No issues identified.

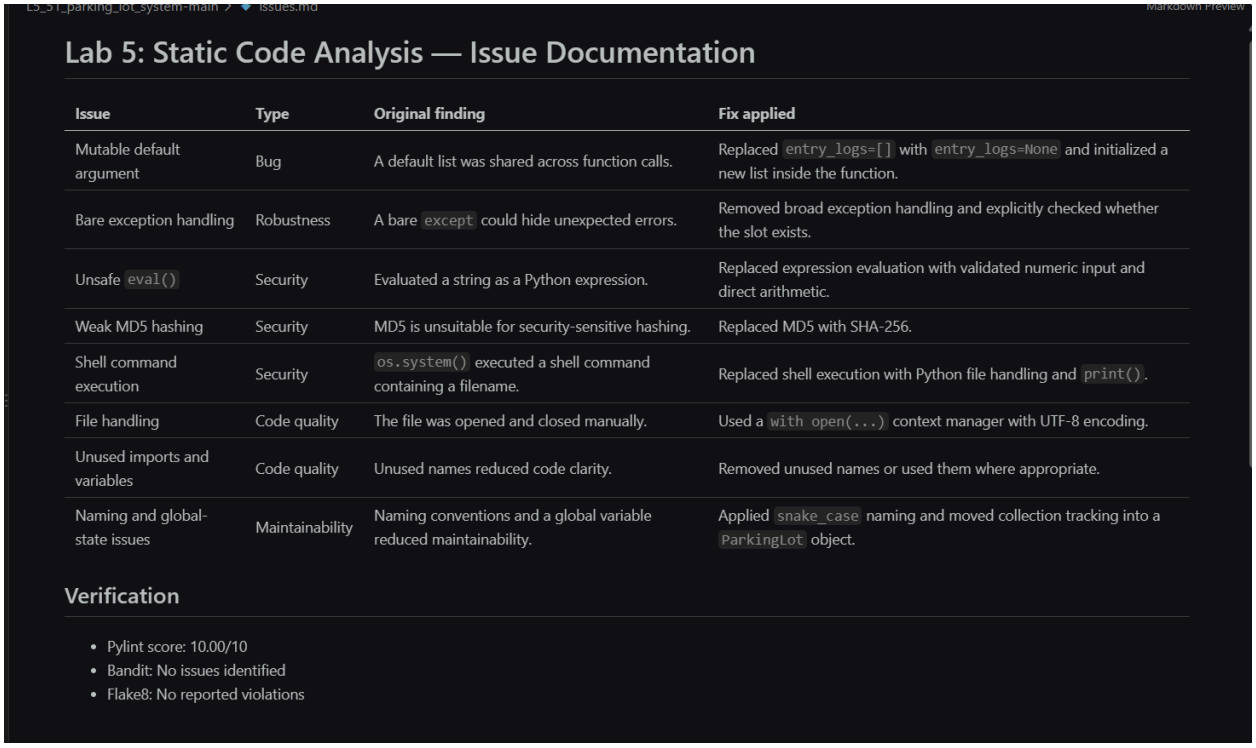
Code scanned:
        Total lines of code: 124
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

3. Flake8 Style Analysis

```
_system-main> python -m flake8 parking_lot_system.py
(venv) PS C:\Users\dell\Downloads\L5_51_parking_lot_system-main\L5_51_parking_lot
```

4. Issue Documentation Table



Lab 5: Static Code Analysis — Issue Documentation

Issue	Type	Original finding	Fix applied
Mutable default argument	Bug	A default list was shared across function calls.	Replaced `entry_logs=[]` with `entry_logs=None` and initialized a new list inside the function.
Bare exception handling	Robustness	A bare `except` could hide unexpected errors.	Removed broad exception handling and explicitly checked whether the slot exists.
Unsafe `eval()`	Security	Evaluated a string as a Python expression.	Replaced expression evaluation with validated numeric input and direct arithmetic.
Weak MD5 hashing	Security	MD5 is unsuitable for security-sensitive hashing.	Replaced MD5 with SHA-256.
Shell command execution	Security	`os.system()` executed a shell command containing a filename.	Replaced shell execution with Python file handling and `print()`.
File handling	Code quality	The file was opened and closed manually.	Used a `with open(...)` context manager with UTF-8 encoding.
Unused imports and variables	Code quality	Unused names reduced code clarity.	Removed unused names or used them where appropriate.
Naming and global-state issues	Maintainability	Naming conventions and a global variable reduced maintainability.	Applied `snake_case` naming and moved collection tracking into a `ParkingLot` object.

Verification

- Pylint score: 10.00/10
- Bandit: No issues identified
- Flake8: No reported violations

5. Reflection Questions

